Math Stuff

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§1 Abstract Algebra

§1.1 Group Theory

Theorem 1.1 (Lagrange's Theorem)

Let G be a finite group and H be a subgroup of G. Then the order of H divides the order of G, and the index $[G : H] = |G|/|H|$ is exactly the number of distinct left cosets of H in G.

Proof. We define an equivalence relation on G by setting $a \sim b$ if and only if $a^{-1}b \in H$. The equivalence classes under this relation are precisely the left cosets of H, which take the form $aH = \{ah \mid h \in H\}$. Because equivalence classes partition the underlying set, G is the disjoint union of the distinct left cosets of H.

Next, we establish that every left coset has the exact same cardinality as H. For any fixed $a \in G$, define the map $f\colon H \to aH$ by $f(h) = ah$. This map is surjective by definition. It is also injective because $ah_1 = ah_2$ implies $h_1 = h_2$ by left cancellation in G. Thus, f is a bijection, and $|aH| = |H|$.

Let $k = [G : H]$ be the number of distinct left cosets. Since G is partitioned into k disjoint sets, each of exactly size $|H|$, it logically follows that $|G| = k|H|$. Therefore, $|H|$ divides $|G|$. $\square$

Theorem 1.2 (First Isomorphism Theorem)

If $f : G \to H$ is a homomorphism, then $\ker f$ is a normal subgroup of G and

$$G/\ker f \cong \operatorname{im} f.$$

Proof. Let $K = \ker f$. We claim that K is a subgroup. It does contain 1 (because $f(1) = 1$); if $x, y \in K$ (so that $f(x) = 1 = f(y)$, then $f(xy) = f(x)f(y) = 1$ and $xy \in K$; if $x \in K$, then $f(x^{-1}) = f(x)^{-1} = 1$ and $x^{-1} \in K$. Furthermore, the subgroup K is normal: if $x \in K$ and $g \in G$, then $f(gxg^{-1}) = f(g)f(x)f(g)^{-1} = f(g)f(g)^{-1}$ and so $gxg^{-1} \in K$.

Define $\phi : G/k \to \operatorname{im} f$ by $\phi(xK) = f(x)$. Now ϕ is well defined: if $x'K = xK$, then $x' = xk$ for some $k \in K$, and $f(x') = f(xk) = f(x)f(k) = f(x)$. It is routine to check that ϕ is homomorphism with $\operatorname{im}\phi = \operatorname{im} f$. Finally, ϕ is an injection, because $\phi(xK) = 1$ implies $f(x) = 1$, hence $x \in K$ and $xK = K$. □

Lemma 1.3

If K and H are subgroups of G with K normal in G, then $K \vee H = KH = \{kh : k \in K \text{ and } h \in H\} = HK$.

Proof. Clearly $KH \subset K \vee H$. For the reverse inclusion, it suffices to prove that KH is a subgroup, for it does contain $K \cup H$. Now $khk_1h_1 = k(hk_1h^{-1})hh_1 = (kk_2)(hh_1) \in KH$ for some $k_2 \in K$(because K is normal). Also $(kh)^{-1} = h^{-1}k^{-1} = (h^{-1}k^{-1}h)h^{-1} = k'h^{-1} \in KH$ for some $k' \in K$(again, because K is normal). Therefore, KH is a subgroup. if $hk \in HK$, then $hk = (hkh^{-1})h = k'h \in KH$ for some $k' \in K$, and so $HKsubsetKH$; the reverse inclusion is proved similarly. □

Theorem 1.4 (Second Isomorphism Theorem)

If K and H are subgroups G with K normal in G with K normal in G, then $K \cap H$ is a normal subgroup of H and

$$H/(K \cap H) \cong KH/K.$$

Proof. Let $\pi : G \to G/K$ be the natural map, defined by $\pi(x) = xK$, and let $f : H \to G/K$ be the restriction $\pi|H$. Now $\ker f = K \cap H$ and $\operatorname{im} f$ is the family of all cosets xK in G/K with $x \in H$ (hence $\operatorname{im} f = KH/K$). The first isomorphism theorem now gives the result. □

Theorem 1.5 (Third Isomorphism Theorem)

Theorem 1.6 (The Orbit-Stabilizer Theorem)

Let G be a finite group acting on a set X. For any $x \in X$, the size of the orbit of x multiplied by the size of its stabilizer subgroup equals the order of the group:

$$|G| = |\mathrm{Orb}(x)| \cdot |\mathrm{Stab}(x)|.$$

Proof. Fix $x \in X$. The stabilizer of x, denoted $H = \mathrm{Stab}(x) = \{g \in G \mid g \cdot x = x\}$, is a subgroup of G. The orbit of x is the set $\mathrm{Orb}(x) = \{g \cdot x \mid g \in G\}$.

We construct a bijection between the left cosets of H in G and the elements of the orbit $\mathrm{Orb}(x)$. Define the mapping $\phi\colon G/H \to \mathrm{Orb}(x)$ by $\phi(gH) = g \cdot x$.

Well-definedness and Injectivity: We must show that ϕ is well-defined and injective, which amounts to proving that $g_1H = g_2H \iff g_1 \cdot x = g_2 \cdot x$. We observe the following chain of equivalences:

$$g_1H = g_2H \iff g_2^{-1}g_1 \in H \iff (g_2^{-1}g_1) \cdot x = x \iff g_1 \cdot x = g_2 \cdot x.$$

This confirms both that the map's output does not depend on the choice of coset representative, and that distinct cosets perfectly map to distinct elements of the orbit.

Surjectivity: For any $y \in \mathrm{Orb}(x)$, there exists some $g \in G$ such that $y = g \cdot x$. By definition, $\phi(gH) = y$, making the map surjective.

Since ϕ is a bijection, the number of left cosets $[G : H]$ is exactly $|\mathrm{Orb}(x)|$. By Lagrange's Theorem, $|G| = [G : H] \cdot |H|$, which yields $|G| = |\mathrm{Orb}(x)| \cdot |\mathrm{Stab}(x)|$. □

Theorem 1.7 (Cauchy's Theorem)

If G is a finite group and p is a prime number dividing $|G|$, then G contains an element of order p.

Proof. We employ McKay's combinatorial proof via group actions. Consider the set of p-tuples of elements from G whose product is the identity:

$$X = \{(g_1, g_2, \ldots, g_p) \in G^p \mid g_1g_2 \ldots g_p = e\}.$$

The choice of the first $p-1$ elements uniquely determines the final element as $g_p = (g_1 \ldots g_{p-1})^{-1}$. Thus, the cardinality of X is exactly $|G|^{p-1}$. Since p divides $|G|$, it follows that p also divides $|X|$.

We define an action of the cyclic group $\mathbb{Z}_p = \langle \sigma \rangle$ on X by cyclic permutation. Specifically, let the generator σ act by shifting the coordinates:

$$\sigma \cdot (g_1, g_2, \ldots, g_p) = (g_2, g_3, \ldots, g_p, g_1).$$

Notice that if $g_1g_2 \ldots g_p = e$, then conjugating by g_1^{-1} yields $g_2 \ldots g_pg_1 = e$. Thus, the action is well-defined and maps X into itself.

By the Orbit-Stabilizer Theorem, the size of every orbit under this $\mathbb{Z}_p$-action must divide $|\mathbb{Z}_p| = p$. Since p is prime, every orbit has size exactly 1 or p. An orbit has size 1 if and only if the tuple is fixed under cyclic shifts, which requires $g_1 = g_2 = \cdots = g_p = g$, and thus $g^p = e$.

Let k be the number of orbits of size 1. The tuple $(e, e, \ldots, e)$ clearly forms an orbit of size 1, so $k \geq 1$. The total number of elements $|X|$ is the sum of the sizes of the

individual orbits. Since $|X|$ is divisible by p, and all other orbits have size exactly p, the number of orbits of size 1 must also be a multiple of p.

Since $k \geq 1$ and p divides k, we must have $k \geq p \geq 2$. Therefore, there exists at least one other tuple $(g, g, \ldots, g)$ with $g \neq e$ such that $g^p = e$. This g is our desired element of order p. □

§1.2 Ring Theory

Theorem 1.8 (Maximal Ideals and Fields)

Let R be a commutative ring with identity. An ideal $M \subset R$ is maximal if and only if the quotient ring R/M is a field.

Proof. By the Correspondence Theorem for Rings, the ideals of R/M are in exact bijection with the ideals of R that contain M.

Necessity: Suppose M is a maximal ideal. By definition, there are no intermediate ideals strictly between M and R. Consequently, under the correspondence bijection, the quotient ring R/M has exactly two ideals: the zero ideal $(0) = M/M$ and the entire ring R/M. A commutative ring with identity possesses exactly two ideals if and only if it is a field. Thus, R/M is a field.

Sufficiency: Suppose R/M is a field. The only ideals of a field are the zero ideal and the field itself. By the correspondence bijection, there are exactly two ideals in R containing M, which must be M itself and R. This dictates that no proper ideal strictly contains M, rendering M maximal. □

Theorem 1.9 (Prime Ideals and Integral Domains)

Let R be a commutative ring with identity. A proper ideal $P \subset R$ is prime if and only if the quotient ring R/P is an integral domain.

Proof. We evaluate the condition for zero-divisors in the quotient ring R/P.

Necessity: Suppose P is a prime ideal. Let $(a + P)$ and $(b + P)$ be elements of R/P such that their product is the zero element of the quotient, meaning $(a + P)(b + P) = ab + P = 0 + P$. This implies that $ab \in P$. By the definition of a prime ideal, either $a \in P$ or $b \in P$. Translating this back to the quotient ring, either $a + P = 0 + P$ or $b + P = 0 + P$. Since there are no non-zero zero-divisors, R/P is an integral domain.

Sufficiency: Suppose R/P is an integral domain. Let $a, b \in R$ such that $ab \in P$. Moving to the quotient ring, this means $(a + P)(b + P) = ab + P = 0 + P$. Because R/P is an integral domain, it has no zero-divisors, so we must have either $a + P = 0 + P$ or $b + P = 0 + P$. This forces either $a \in P$ or $b \in P$, exactly satisfying the definition of a prime ideal. □

Theorem 1.10 (The Chinese Remainder Theorem for Rings)

Let R be a commutative ring with identity, and let $I_1, I_2, \ldots, I_n$ be pairwise comaximal ideals (i.e., $I_i + I_j = R$ for all $i \neq j$). Then the natural map

$$\phi\colon R \to R/I_1 \times R/I_2 \times \cdots \times R/I_n$$

defined by $\phi(r) = (r + I_1, r + I_2, \ldots, r + I_n)$ is a surjective ring homomorphism with kernel $\bigcap_{i=1}^n I_i$. Consequently, $R/\bigcap_{i=1}^n I_i \cong \prod_{i=1}^n R/I_i$.

Proof. The map ϕ is clearly a ring homomorphism by component-wise evaluation. Its kernel consists of all elements $r \in R$ such that $r \in I_i$ for every i, giving $\ker \phi = \bigcap_{i=1}^n I_i$. The heavy lifting of the proof is demonstrating surjectivity.

We first prove the case for $n = 2$. Since I_1 and I_2 are comaximal, there exist elements $x \in I_1$ and $y \in I_2$ such that $x + y = 1$. Let $(a + I_1, b + I_2)$ be an arbitrary element in $R/I_1 \times R/I_2$. Consider the element $r = ay + bx$. Modulo I_1, we observe $x \equiv 0$, yielding $r \equiv ay \equiv a(1 - x) \equiv a \pmod{I_1}$. Modulo I_2, we observe $y \equiv 0$, yielding $r \equiv bx \equiv b(1 - y) \equiv b \pmod{I_2}$. Thus, $\phi(r) = (a + I_1, b + I_2)$, proving surjectivity for $n = 2$.

For $n > 2$, we proceed by constructing elements $e_i \in R$ such that $e_i \equiv 1 \pmod{I_i}$ and $e_i \equiv 0 \pmod{I_j}$ for all $j \neq i$. Fix i. For each $j \neq i$, we have $I_i + I_j = R$, so we can find $x_j \in I_i$ and $y_j \in I_j$ such that $x_j + y_j = 1$.

Taking the product over all $j \neq i$, we get $\prod_{j \neq i}(x_j + y_j) = 1$. Expanding this product, every term contains at least one x_j (and thus belongs to I_i) except the single term $\prod y_j$. Let $e_i = \prod_{j \neq i} y_j$. Then $e_i \in \bigcap_{j \neq i} I_j$, forcing $e_i \equiv 0 \pmod{I_j}$ for $j \neq i$. Moreover, $1 - e_i \in I_i$, so $e_i \equiv 1 \pmod{I_i}$.

Given any arbitrary target $(r_1 + I_1, \ldots, r_n + I_n)$, the element $r = r_1 e_1 + \cdots + r_n e_n$ satisfies $\phi(r) = (r_1 + I_1, \ldots, r_n + I_n)$. Thus, ϕ is surjective. The result then follows immediately from the First Isomorphism Theorem. □

§2 Graph Theory & Combinatorics

Lemma 2.1 (The Handshaking Lemma)

Let $G = (V, E)$ be a finite undirected graph. Then the sum of the degrees of all vertices is twice the number of edges:

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof. We proceed by the method of double counting. Consider the set of incidence pairs (v, e), where $v \in V$ is a vertex incident to the edge $e \in E$.

On one hand, if we group these pairs by vertices, each vertex v participates in exactly $\deg(v)$ such pairs, yielding a total of $\sum_{v \in V} \deg(v)$ pairs. On the other hand, if we group them by edges, each edge $e \in E$ has exactly two endpoints, and thus participates in exactly two pairs. Summing over all edges yields $2|E|$ pairs. Equating the two counts yields the result. □

Theorem 2.2 (Ramsey's Theorem for $R(3,3)$)

In any group of 6 people, there is always either a group of 3 mutual friends or a group of 3 mutual strangers. Equivalently, any 2-coloring of the edges of the complete graph K_6 contains a monochromatic triangle.

Proof. Let the vertices of K_6 represent the people, and let the edges be colored red (friends) or blue (strangers). Choose an arbitrary vertex v. Since the degree of v is 5, the Pigeonhole Principle dictates that at least $\lceil 5/2 \rceil = 3$ edges incident to v must share the same color. Without loss of generality, assume there are 3 red edges connecting v to vertices A, B, and C.

Now consider the triangle formed by A, B, and C. If any single edge of this triangle (say AB) is red, then the vertices v, A, and B form a monochromatic red triangle. Alternatively, if no edges of the triangle ABC are red, then they must all be blue, meaning A, B, and C form a monochromatic blue triangle. In either case, a monochromatic triangle is forced. □

Theorem 2.3 (Euler's Theorem for Eulerian Graphs)

A finite, connected, undirected graph G contains an Eulerian circuit if and only if every vertex in G has an even degree.

Proof. We must establish both necessity and sufficiency.

Necessity: The forward implication is an elegant consequence of parity. Suppose G possesses an Eulerian circuit, which by definition traverses every edge exactly once and returns to its starting vertex. As we trace this circuit, every time we enter a vertex v, we must also inevitably exit it. Thus, each visit to v consumes exactly two incident edges (one in, one out). Since every edge incident to v is traversed exactly once during the entire circuit, the total number of edges incident to v must be a multiple of two. Hence, $\deg(v)$ is even for all $v \in V$.

Sufficiency: For the reverse direction, we proceed by induction on the number of edges $|E|$. The base case is trivial. Assume the statement holds for all connected graphs with strictly fewer than $|E|$ edges.

Since G is connected and every vertex has an even degree (meaning at least degree 2), G cannot be a tree and must contain a cycle. Start at an arbitrary vertex u and walk along unvisited edges. Because every vertex has an even degree, we can never get "stuck" at a vertex other than our starting vertex u; every time we enter a new vertex, there is guaranteed to be an unvisited edge leaving it. This walk must eventually close upon itself, forming a cycle C.

If C contains all the edges of G, the proof is complete. If not, we remove the edges of C from G. The resulting graph, $G \setminus C$, may break into several disconnected components. However, because we removed exactly two edges from every vertex involved in C, every vertex in $G \setminus C$ still has an even degree.

By our inductive hypothesis, each non-trivial connected component of $G \setminus C$ possesses its own Eulerian circuit. Furthermore, because the original graph G was connected, each of these components must share at least one vertex with the cycle C.

To construct the Eulerian circuit for the entirety of G, we begin traversing C. Whenever we reach a vertex that acts as an intersection point with a component of $G \setminus C$, we pause our walk on C, "splice in" the Eulerian circuit of that component, traverse it completely,

and then resume our walk along C. This combined journey traverses every edge of G exactly once, completing the proof. □

§3 Real Analysis

Theorem 3.1 (Banach Fixed Point Theorem)

Let (X, d) be a non-empty complete metric space and let $T\colon X \to X$ be a contraction mapping; that is, there exists a constant $c \in [0, 1)$ such that $d(T(x), T(y)) \leq c \cdot d(x, y)$ for all $x, y \in X$. Then T admits a unique fixed point $x^* \in X$.

Proof. We first establish existence. Choose an arbitrary $x_0 \in X$ and define a sequence by $x_{n+1} = T(x_n)$. We claim (x_n) is a Cauchy sequence. By induction on the contraction property, the distance between consecutive terms is bounded by $d(x_n, x_{n+1}) \leq c^n d(x_0, x_1)$. For any $m > n$, we use the triangle inequality and the geometric series to bound $d(x_n, x_m)$:

$$d(x_n, x_m) \leq \sum_{i=n}^{m-1} d(x_i, x_{i+1}) \leq d(x_0, x_1) \sum_{i=n}^{m-1} c^i \leq d(x_0, x_1) \frac{c^n}{1-c}.$$

Since $c < 1$, the term $c^n \to 0$ as $n \to \infty$. Thus, (x_n) is Cauchy. Because X is complete, the sequence converges to some limit $x^* \in X$. Since T is a contraction, it is uniformly continuous, so $T(x^*) = T(\lim x_n) = \lim T(x_n) = \lim x_{n+1} = x^*$. Thus x^* is a fixed point.

For uniqueness, suppose there exists another fixed point y^*. Then $d(x^*, y^*) = d(T(x^*), T(y^*)) \leq c \cdot d(x^*, y^*)$. Since $c < 1$, this inequality can only hold if $d(x^*, y^*) = 0$, meaning $x^* = y^*$. □

§4 Complex Analysis

Theorem 4.1 (Liouville's Theorem)

Every bounded entire function is constant.

Proof. Let $f\colon \mathbb{C} \to \mathbb{C}$ be an entire function (holomorphic everywhere), and suppose there exists $M > 0$ such that $|f(z)| \leq M$ for all $z \in \mathbb{C}$. Fix any $z_0 \in \mathbb{C}$ and let $R > 0$ be the radius of a circle C_R centered at z_0. By Cauchy's Integral Formula for derivatives, we have

$$f'(z_0) = \frac{1}{2\pi i} \oint_{C_R} \frac{f(z)}{(z - z_0)^2}\, dz.$$

We apply the standard estimation lemma (the ML-inequality). The length of C_R is $2\pi R$, and on the curve, $|z - z_0| = R$. Thus,

$$|f'(z_0)| \leq \frac{1}{2\pi} \cdot \frac{M}{R^2} \cdot 2\pi R = \frac{M}{R}.$$

Since f is entire, this inequality holds for any arbitrary $R > 0$. Taking the limit as $R \to \infty$, we see that $|f'(z_0)| \leq 0$, which forces $f'(z_0) = 0$. Since z_0 was arbitrary, the derivative is zero everywhere on $\mathbb{C}$, implying f is a constant function. □

§5 Numerical Analysis

Theorem 5.1 (Polynomial Interpolation)

Given $n+1$ distinct points $(x_0, y_0), \ldots, (x_n, y_n)$ in $\mathbb{R}^2$, there exists a unique polynomial $P(x)$ of degree at most n such that $P(x_i) = y_i$ for all $i = 0, \ldots, n$.

Proof. **Existence (Lagrange Construction):** We construct the polynomial explicitly. Define the Lagrange basis polynomials $\ell_i(x)$ for $0 \leq i \leq n$ as:

$$\ell_i(x) = \prod_{0 \leq j \leq n, j \neq i} \frac{x - x_j}{x_i - x_j}.$$

Notice that $\ell_i(x_i) = 1$ and $\ell_i(x_j) = 0$ for $i \neq j$. Furthermore, each $\ell_i(x)$ is a product of n linear terms, so its degree is exactly n. The desired polynomial is the linear combination:

$$P(x) = \sum_{i=0}^{n} y_i \ell_i(x).$$

By construction, $P(x_i) = y_i$ for all i, and $\deg(P) \leq n$.

Uniqueness: Suppose there is another polynomial $Q(x)$ of degree at most n satisfying $Q(x_i) = y_i$. Consider the difference polynomial $D(x) = P(x) - Q(x)$. The degree of $D(x)$ is at most n. However, $D(x_i) = P(x_i) - Q(x_i) = 0$ for all $n+1$ points $x_0, \ldots, x_n$. The Fundamental Theorem of Algebra states that a non-zero polynomial of degree n can have at most n roots. Since $D(x)$ has $n+1$ roots, it must be the identically zero polynomial, forcing $P(x) = Q(x)$. □

§6 Mathematical Statistics

Theorem 6.1 (Central Limit Theorem)

Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables with expected value μ and finite variance $\sigma^2 > 0$. Then the sequence of random variables

$$Z_n = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to a standard normal random variable $\mathcal{N}(0,1)$ as $n \to \infty$.

Proof. Without loss of generality, we may center and scale the random variables by assuming $\mu = 0$ and $\sigma = 1$ (by replacing X_i with $(X_i - \mu)/\sigma$). Thus, $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i$.

Let $M_X(t)$ be the moment generating function (MGF) of X_i. By Taylor's Theorem, expanding $M_X(t)$ around $t = 0$ gives:

$$M_X(t) = M_X(0) + tM_X'(0) + \frac{t^2}{2} M_X''(0) + \mathcal{O}(t^3) = 1 + 0 + \frac{t^2}{2} + \mathcal{O}(t^3).$$

By the properties of MGFs for independent variables, the MGF of Z_n is:

$$M_{Z_n}(t) = \left(M_X\left(\frac{t}{\sqrt{n}}\right)\right)^n = \left(1 + \frac{t^2}{2n} + \mathcal{O}\left(\frac{t^3}{n^{3/2}}\right)\right)^n.$$

Taking the limit as $n \to \infty$, we invoke the standard calculus limit $\lim_{n\to\infty}(1+x/n)^n = e^x$. The higher-order error terms vanish faster than $1/n$, yielding:

$$\lim_{n\to\infty} M_{Z_n}(t) = e^{t^2/2}.$$

This limit is precisely the moment generating function of the standard normal distribution $\mathcal{N}(0,1)$. Since MGFs uniquely determine distributions, $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. □

§7 Linear Algebra

Theorem 7.1 (The Rank-Nullity Theorem)

Let V and W be vector spaces over a field, with V finite-dimensional. Let $T\colon V \to W$ be a linear transformation. Then

$$\dim(V) = \dim(\ker T) + \dim(\operatorname{im} T).$$

Proof. Let $n = \dim(V)$ and $k = \dim(\ker T)$. Since $\ker T$ is a subspace of V, we can choose a basis $\{v_1, \ldots, v_k\}$ for $\ker T$. By the Basis Extension Theorem, we can append $m = n - k$ linearly independent vectors to form a full basis for V:

$$\mathcal{B}_V = \{v_1, \ldots, v_k, u_1, \ldots, u_m\}.$$

We claim that the set $S = \{T(u_1), \ldots, T(u_m)\}$ forms a basis for $\operatorname{im} T$.

First, we show S spans $\operatorname{im} T$. For any $w \in \operatorname{im} T$, there exists $v \in V$ such that $T(v) = w$. Writing v as a linear combination of $\mathcal{B}_V$, we have $v = \sum c_i v_i + \sum d_j u_j$. Applying the linear map T, and noting that $T(v_i) = 0$ because they are in the kernel, we obtain:

$$w = T(v) = \sum c_i T(v_i) + \sum d_j T(u_j) = \sum d_j T(u_j).$$

Thus, S spans the image.

Next, we verify linear independence. Suppose $\sum a_j T(u_j) = 0$. By linearity, this means $T(\sum a_j u_j) = 0$. This implies the vector $x = \sum a_j u_j$ lies in $\ker T$. Thus, x can be written in terms of the kernel's basis: $x = \sum b_i v_i$. Rearranging gives:

$$\sum a_j u_j - \sum b_i v_i = 0.$$

Because $\mathcal{B}_V$ is a basis for V, all coefficients in this linear combination must be zero. Specifically, $a_j = 0$ for all j, proving S is linearly independent.

Therefore, S is a basis for $\operatorname{im} T$, meaning $\dim(\operatorname{im} T) = m = n - k = \dim(V) - \dim(\ker T)$. □